Lab#6



– Digital Electronics –

D/A Conversion

The purpose of this laboratory session is to use a simple NIOS-based system and some custom logic to build a D/A converter

Abbreviations and acronyms:
IC – Integrated Circuit
LED – Light Emitting Diode
MUX – Multiplexer
]

USE EXACTLY THE SAME I/O PINS SPECIFIED IN THIS DOCUMENT.

Introduction and theory

There are many possibilities for D/A conversion, that is transforming a digital value, a number into an electrical value, voltage or current. The most attractive ones under several points of view are those obtained through the Pulse Width Modulation (PWM) and the PPM (Pulse Proportion Modulation), also called PDM (Pulse Density Modulation). In PWM the duty cycle (D) of a fixed frequency digital waveform takes values in the range 0 N. Besides, D is made proportional to a digital value, while in PPM the number of pulses of fixed duration over a defined time interval is proportional to the digital value. In both cases it is very simple to show that the average value of the output waveform is proportional to the digital value and that the analog signal equal to it may be obtained by a low pass filter, but the filter requirements are different for the two solutions.

1. The PWM solution

This is the simplest from the implementation point of view. Let us assume for simplicity that the digital number range is between 0 and N (0÷N-1). In a straightforward implementation we can use a periodic clock frequency f_0 and two counters, the first initialized to the number D to be converted (with D ϵ [0, N-1]) and the second one to N-1. We set the output to 1 (V_{dd}) and we decrement both counters every clock cycle. When the first one reaches 0 we set the output to 0 and we continue decrementing the second one. When also the second counter reaches 0, then we reload the counters with the original values and we start again the process.

The filter must be designed in order to reject all harmonic components and transfer only the average value. The value of the DC component is V_{dd} D/N, whereas, the first harmonic component is at frequency f_0/N and its amplitude depends on the duty cycle as $V_{dd}/\pi \sin(\pi D/N)$. It is maximum when the duty cycle is 50% and its amplitude is V_{dd}/π. The attenuation of the first harmonic must be at least lower than half of the minimum value to be converted.

2. The PPM/PDM solution

The basis for the PPM/PDM solution is the so called Sigma-Delta modulator. Figure 1 shows the analog to PPM/PDM converter. The 1-bit DAC will output an analog value equal either to the maximum or minimum value. At each clock pulse the output of the DAC will be added to the signal, integrated and compared to a threshold value. Without entering into the theory of sigma-delta converters, just note that due to the negative feedback loop the average output level at the 1-bit DAC must always be equal to the input signal level. The output of the converter is a bit stream whose average value is equal to the output signal value.

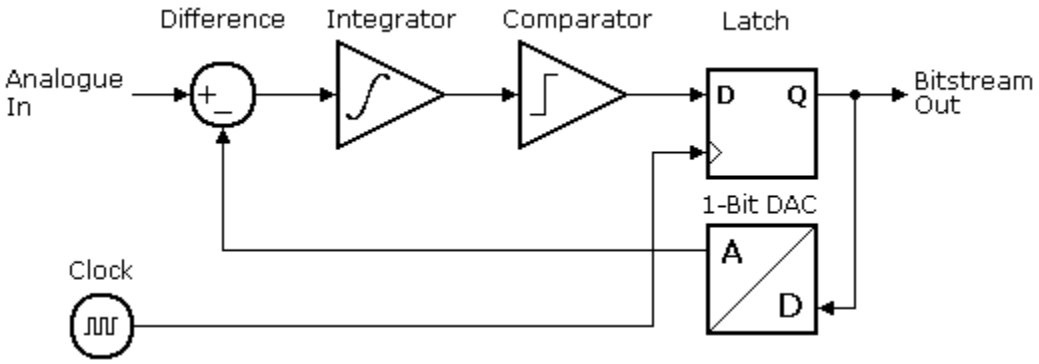


Figure 1 – Analog PPM converter

The digital counterpart of this modulator is the digital sigma-delta modulator, which is shown in Figure 2. Here the integrator is substituted by an adder and an accumulator register and the 1-bit DAC by 1-bit DDC that has as output equal to the maximum or minimum input value according to the output of the comparator. Since the value of the threshold is not important, we can implement the comparator just by selecting the Most Significant Bit of the accumulator register.

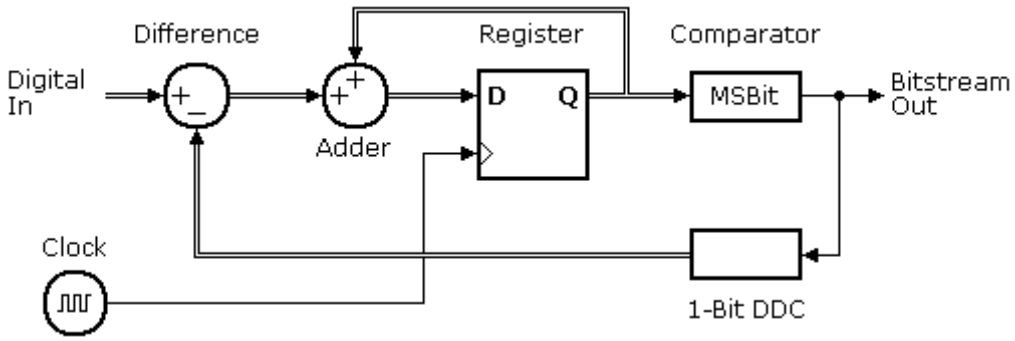


Figure 2 – Digital PPM converter

The only important issue is the sizing of both the adder and the accumulator register in order to avoid overflow. Both should be sized at least two bits larger than the input and the DDC output.

The analysis of the requirements of the filter is more complex, but it is intuitive that the PPM/PDM waveform has more energy at the higher harmonics and therefore less stringent requirements on the filter.
A first order evaluation shows that the filter requires an attenuation of the same order of magnitude of that of the previous case, but at frequency f_0 instead of at frequency f_0/N.

Note: while PWM assumes the input value to be transformed into the duty cycle is positive, PPM/PDM assumes the digital input is either positive or negative.

Assignment

1) **Design** and describe in HDL both the PWM and PPM/PDM D/A converters (8 input bits) and test them with a proper testbench. Each converter receives 50 MHz as the clock frequency and has also a two input port to specify the value of frequency f_0 according to the following table:

   0 0 400 kHz
   0 1 200 kHz
   1 0 100 kHz
   1 1 50 kHz

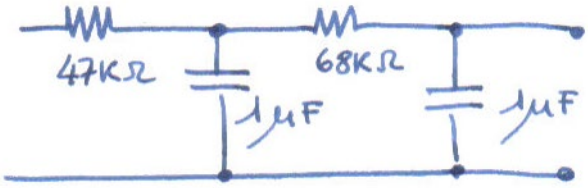
2) **Create** a NIOS system which includes a timer peripheral and two PIOs: an 8 bit output PIO and a 2 bit output PIO. Connect the 8 bit output PIO as the input value of the two converters (PWM and PPM/PDM) and to two 7-segment displays. Connect the 2 bit PIO to two LEDs and to frequency selection input of the two converters.

3) **Write** a first C-program which i) reads from the host-computer keyboard an 8-bit value (the value to convert) and the value of the frequency (according with the previous table); ii) outputs these values to the custom hardware.

4) **Write** a second C-program which contains:

   - An initialization routine to configure and enable the timer for working in interrupt. Set the timer interval to an equivalent frequency of 100 Hz.
   - An interrupt routine which (at each timer interrupt) sends to the PWM and the PPM/PDM peripherals a value from an array of 20 values (file *sinus.h*). This array contains a sampled sinusoidal waveform with an added offset equal to 127. The array must be scanned cyclically and when the last value is reached the scan must restart from the first element.

5) **Evaluate** the frequency response of the low pass filter shown in the following figure. Then, **connect** the output of each converter to the filter.



6) **Run** the first program (point 3) with output data values equal to 20, 128 and 160 for all four values of f_0. Then, for each configuration, measure the average value of the output and the ripple at the output of each converter. Discuss the results (after evaluating the frequency response of the filter).

7) **Run** the second program (point 4), which enables the timer, and **measure** the amplitude of the sinusoidal waveform generated by the PWM and the PPM/PDM converters after having

changed the capacitors of the filter from 1 μF to 47nF. Increase gradually the timer interval up to an equivalent frequency equal to 1 kHz and describe the effects on the PPM/PDM and PWM converters using the 4 different clock frequencies.

Deliverable report must contain:

- A snapshot of the processor configuration of the SOPC tool.
- The block scheme of the system.
- The top entity HDL description.
- The HDL descriptions of the PWM and the PPM converters.
- The complete listing of the programs, with comments to explain the code.
- A report about the measurements and a discussion of the results.